



राष्ट्रीय भू-सूचना विज्ञान
एवं प्रौद्योगिकी संस्थान
भारतीय सर्वेक्षण विभाग
विज्ञान और प्रौद्योगिकी विभाग

National Institute for Geo-Informatics
Science & Technology
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Department of Science & Technology

GNSS Instrument Specifications

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What is a GNSS Receiver?



A GNSS (Global Navigation Satellite System) receiver is an electronic device that receives, processes, and interprets signals transmitted by navigation satellites such as GPS, GLONASS, Galileo and BeiDou.

The receiver computes Position, Velocity, and Time (PVT) by decoding extremely weak satellite signals that travel over 20,000 km to reach Earth. Modern GNSS receivers are capable of multi-constellation tracking, multi-frequency measurements, and correction-based enhanced positioning such as DGPS, RTK, and PPP.

Is my receiver capable to receive signals of all constellations in all the frequency bands?

Really do we need to use all the satellites visible in the sky?

Based on your survey accuracy requirement selection of receivers is to be done. Different types of receivers which can give us different types of solutions.

Regarding Error mitigation, whether my receiver is capable to resolve the problem?

Data rate from 1 Hz i.e., 1 solution per second to nowadays can go upto 20/50/100 hz per second.

Capability of a receiver of using multiple frequencies.

Whether your receiver has all these features or not, whether all the GNSS positioning techniques can be taken up using the receiver which you are going to procure?

Supported Constellations and Frequencies

GNSS receivers can be categorized based on the type of signals used and the application. GNSS Frequency Types, Codes, and Usage:

- Primary Civil Frequency - L1 / E1 / B1 (≈ 1575 MHz)
- Dual Frequency & Surveying - L2 / G2 / B2 (≈ 1227 – 1246 MHz)
- Aviation & Safety-of-Life - L5 / E5a / B2a (≈ 1176 MHz)
- High Accuracy / Multipath Resistant E5b (≈ 1207 MHz)
- Combined Wideband Signal - E5ab AltBOC
- Commercial / Encrypted Services E6 / B3 (≈ 1278 – 1268 MHz)
- GLONASS Modern CDMA Signals (New Standard)

L1 / E1 / B1 (≈1575 MHz) – Used for: Standard positioning, Navigation for all receivers, Fast acquisition, Single-frequency GNSS.

Signal/Codes are **GPS** L1: C/A, P(Y), L1C (modern civil) **Galileo** E1: E1-B, E1-C (Open Service) **BeiDou** B1I (Legacy signal), B1C (Modern Global Civil Signal) **GLONASS** L1OF: C/A, P (FDMA) **GLONASS** L1OC: CDMA modern code, Code Division Multiple Access.

L2 / G2 / B2 (≈1227–1246 MHz) – Used for: Dual-frequency ionospheric correction RTK, Static & PPK. High-precision geodetic surveying.

Signal/Codes are **GPS** L2: P(Y), L2C (civil) **BeiDou** B2I, B2a **GLONASS** L2OF: C/A, P (FDMA) Frequency Division Multiple Access **GLONASS** L2OC: Modern CDMA.

L5 / E5a / B2a (≈1176 MHz) – Used for: Aviation-certified navigation. High-precision PPP/RTK. Strong interference resistance

Signals / Codes: **GPS** L5: L5-I, L5-Q **Galileo** E5a: I/Q pair **Bei Dou** B2a: I/Q

E5b (≈1207 MHz) – Used for: Precision surveying. PPP and advanced navigation. Improved anti-multipath performance

Signals / Codes: **Galileo** E5b: I/Q **BeiDou** B2 (interoperable)

E5ab AltBOC Highest accuracy among GNSS signals, Excellent multipath & noise performance,
Usage: Aviation & high-precision GNSS

Commercial / Encrypted Services E6 / B3 (≈1278–1268 MHz) Used for: Commercial high-precision services
Encrypted/authorized signals. Integrity & authentication. **Galileo** E6: E6-B (commercial),
E6-C (encrypted) **BeiDou** B3I, B3Q

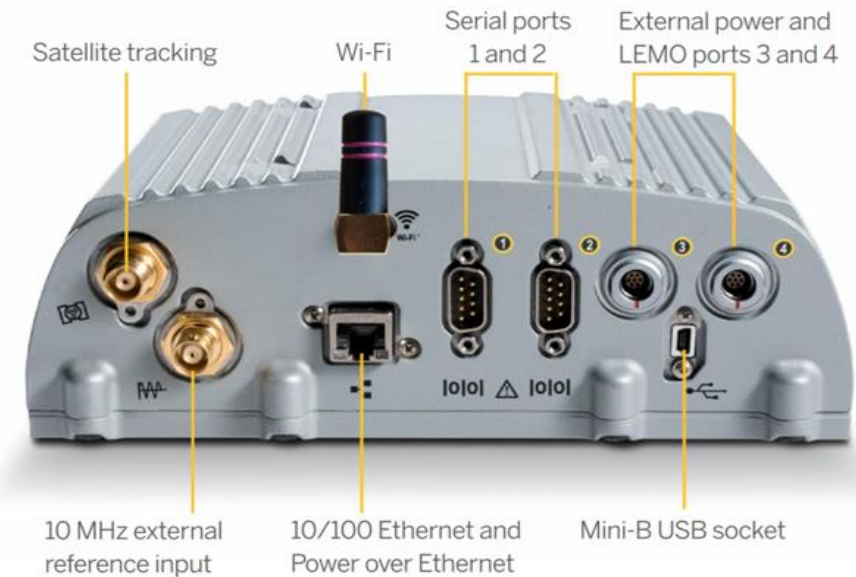
GLONASS Modern CDMA Signals (New Standard) Usage: Interoperability with GPS/Galileo.
Better tracking & anti-jam performance. Future multi-GNSS integration
Signals / Codes: L1OC (1575 MHz) – CDMA L3OC (1207 MHz) Future L5-equivalent
signals

GNSS Receiver Specifications

Specifications typically include:

- Supported Constellations (GPS, GLONASS, Galileo, BeiDou)
- Supported Frequencies (L1/L2/L5/ G1/G2/E1/E5/E6/B1/B2/B3/L5/S-Band)
- Number of Channels (total tracking capability per signal) 672 tracking channels
- Signal Processing Features (DLL/PLL, multipath mitigation, anti-jamming)
- Raw Measurements (Code, Carrier Phase, Doppler, C/N_0)
- Positioning Modes (Single point, DGPS/DGNSS, RTK, PPK, PPP)
- Accuracy Specifications (Static, RTK, PPP accuracy)
- Receiver Output Formats (NMEA, RINEX, RTCM, binary protocols)
- Update Rate / Data Rate (Hz: 1 Hz, 10 Hz, 20 Hz, 50 Hz, etc.)
- Hardware Specs (memory, IMU/tilt sensor, processor type, battery life)
- Environmental Specs (IP rating, temperature, shock, vibration)

Representative GNSS Receiver Specs (Trimble Alloy):



Trimble Alloy – GNSS Receiver Specifications

- Supported Constellations

GPS, GLONASS, Galileo, BeiDou, NavIC, QZSS

Supports all current and planned signals for reference-station use.

- Supported Frequencies

Multi-frequency tracking: L1, L2, L5 (GPS), G1/G2 (GLONASS), E1/E5/E6 (Galileo), B1/B2/B3 (BeiDou), L5/S-band (IRNSS).

Designed for full modern GNSS interoperability.

- Number of Channels

672 tracking channels (dual Maxwell-7 chipsets).

Tracks every signal from every visible satellite.

- Trimble Alloy Accuracy

It varies by correction method

➤ In RTK 8mm+1ppm(Horizontal) & 15mm+1ppm (vertical)

➤ In Static 3mm + 0.1ppm (Horizontal) & 3.5mm + 0.4 ppm (vertical)

- Signal Processing Features
DLL/PLL/FLL tracking loops.
Trimble EVEREST Plus™ multipath mitigation.
Anti-jamming / interference detection, built-in spectrum analyzer.
High signal sensitivity, low-elevation tracking.
- Raw Measurements
Outputs code, carrier phase, Doppler, C/N_0 .
Carrier-phase precision: <1 mm in 1 Hz bandwidth.
Raw data formats: .T02, .T04, BINEX, RINEX 2.x/3.x.
- Positioning Modes
CORS / Permanent Reference Station
RTK Base Station, Campaign / Control point surveying
PPP/RTX support for absolute real-time accuracy
Supports Network RTK (VRS, FKP, MAC).
- Accuracy Specifications
Static / control work: Millimeter-level capability (carrier-phase <1 mm).
Real-time (RTX/RTK): Centimeter-level accuracy.
Designed for high-precision GNSS reference networks.

- Receiver Output Formats

NMEA-0183, RTCM 2.x/3.x, CMR/CMRx, BINEX, GSOF.

Data logging up to 100 Hz; combined data up to 187 Hz.

Output update rate: 20 Hz standard, 50/100 Hz optional.

- Update Rate / Data Rate

Position output: 20 Hz → 50 Hz → 100 Hz (optional).

High-speed raw logging for deformation monitoring.

- Hardware Specifications

Weight: 1.75 kg (with batteries).

Dual hot-swappable Li-ion batteries (~15 hrs operation).

Internal memory: 8 GB, expandable to 24 GB.

Communications: Ethernet, Wi-Fi, Bluetooth, serial ports.

Rugged aluminum chassis.

- Environmental Specifications

IP68 (dust-tight, waterproof immersion).

Operating temperature: –40°C to +65°C.

Shock: 40 g (operating), 75 g (non-operating).

Vibration: MIL-STD-810G compliant.

Signal Quality Parameters

The quality of satellite signal reception is measured using two key parameters:

1) **Signal-to-Noise Ratio (SNR):**

SNR is the ratio of useful signal power (S) to background noise power (N).

It tells us how strong the satellite signal is compared to noise.

Unit is dB

SNR (dB)	Signal Quality
> 40 dB	Very good
30–40 dB	Good
20–30 dB	Weak / noisy
< 20 dB	Very poor, may lose lock

Signal Quality Parameters

-Continued

2) Carrier-to-Noise Density Ratio (C/N_0)

Most important GNSS tracking indicator

Represents the **strength and quality of a GNSS signal** received from a satellite.

Unit is dB-Hz

C/ N_0 (dB-Hz)	Signal Quality	Meaning
> 45 dB-Hz	Excellent	Open sky, very strong tracking
40 – 45 dB-Hz	Good	Normal GNSS conditions
35 – 40 dB-Hz	Moderate	Weakening signal
< 35 dB-Hz	Poor	Obstructions/multipath likely
< 30 dB-Hz	Very weak	Unreliable tracking

Measurements Generated by GNSS Receiver

A GNSS receiver generates the following core measurements:

- 1) **Pseudorange (Code Measurement) — DLL (Delay Lock Loop)** A DLL is an internal tracking loop in a GNSS receiver that is used to track the PRN Code (Pseudo-range) from the satellite. Each satellite transmits a unique code (PRN sequence). The receiver generates an identical replica of that code and compares it with the incoming satellite signal. This is called Code-Correlation. DLL adjusts the time delay until the locally generated code matches with the received code.
- 2) **Carrier Phase Measurement — PLL (Phase Lock Loop)** It is also an internal tracking system in a GNSS receiver that LOCKS & continuously tracks the PHASE of the carrier signal transmitted by a satellite. The GNSS receiver generates a replica of the satellite's carrier signal. The PLL compares the replica with the incoming signal and adjusts itself until both are phase-aligned (Locked).
- 3) **Doppler Measurement** it determines the relative velocity between the satellite and the receiver by observing the frequency shift of the carrier signal. As satellites move, the frequency of the received signal changes (Doppler shift). The GNSS receiver measures this frequency change.

4) **Navigation message** it is the information transmitted by each GNSS satellite that allows the receiver to determine satellite position, timing, and health so that it can compute user position. GNSS receiver demodulates and decodes the data to obtain navigation information

Data Epoch in GNSS Receiver

A data epoch is a specific instant of time at which the GNSS receiver records measurements from all tracked satellite signals.

It is a **time-stamped snapshot** of GNSS observations taken simultaneously from multiple satellites.

At each epoch, the receiver stores:

Measurement	Source
Pseudorange (code)	DLL
Carrier phase	PLL
Doppler	Frequency shift
C/N ₀ (signal quality)	Signal strength
Navigation data	Ephemeris, clock, health, etc.
Time tag	Precise GNSS time (GPS time, Galileo time, etc.)

Data Formats

GNSS receivers output data in various standardized formats for different applications and processing requirements

- 1) **RINEX (Receiver Independent Exchange Format)** - is a standardized, text-based file format used to store raw GNSS observation data.

Observation_Rinex_v3.01

[Observation_Rinex_v3.01.html](#)

Navigation_Rinex_v3.01

[SBAS_Navigation_Rinex_v3.01.html](#)

- 2) **Binary raw observation format** - A binary format is a compact, machine-readable data format used by GNSS receivers to store or transmit GNSS measurements and configuration messages more efficiently than text formats like NMEA or RINEX example .T02, .T04 etc.

Data Formats

-Continued

NMEA (National Marine Electronics Association (NMEA) →

NMEA is a standard ASCII text output describing position, time, satellites, velocity, etc. GNSS receivers generate NMEA data internally and send it out through one of their communication ports like Serial port-232, USB port, Bluetooth, Wi-fi/TCP/IP etc, to external devices. It is a non correction data format. This is the international standard for GNSS position data. This message type is supported by nearly every GNSS receiver on the market. This is a one-way data stream from the GNSS receiver to the device using the data.

RTCM (Radio Technical Commission for Maritime Services) →

It is an international standardized binary format used for transmitting real-time GNSS correction data. RTCM data is **transmitted in real-time**, allowing for immediate correction and high-precision positioning. RTCM standards are used in many areas, including: Communication and navigation equipment, Differential Global Navigation Satellite Systems, Electronic Chart Systems, and Satellite positioning and navigation systems.

Antenna Selection and Its Impact on GNSS Performance

- GNSS antennas act as **spatial and frequency filters**, influencing signal quality.
- Choosing the **correct antenna** is essential for optimal system performance.
- The antenna must match the **receiver's capabilities** and supported frequencies.
- Selection should also consider **size, weight, environmental durability, and mechanical requirements** for the intended application.

Important Parameters for Choosing a GNSS Antenna

- **CONSTELLATION AND SIGNALS –** The antenna must support all required all required signal frequencies and the bandwidths.
- **ANTENNA GAIN –** Minimum antenna gain is necessary to achieve the required carrier-to-noise density ratio (C/No) for reliable satellite tracking. Gain is an important performance indicator of a GNSS antenna, representing its ability to direct or concentrate radio frequency energy in a specific direction. A minimum antenna gain is necessary to achieve the required carrier-to-noise density ratio (C/No) for reliable satellite tracking. Because antenna gain directly affects the C/No observed by the GNSS receiver, it plays a key role in determining the overall tracking performance of the system. Higher gain enables better reception of weak or low-elevation satellite signals, thereby improving the receiver's ability to maintain continuous and accurate tracking.

ELEMENT GAIN – Element Gain determines how effectively the antenna captures the GNSS signals before any amplification or processing occurs. The element gain represents how efficiently an antenna element receives incoming signals. In any RF signal chain, overall performance is limited by its weakest component, so low element gain may be compensated by increasing the gain of the low-noise amplifier (LNA). However, this does not fully recover the signal quality. Although the total gain appears higher, the signal-to-noise ratio (C/No) is still degraded, because noise is amplified along with the signal. Thus, good element gain is essential for maintaining high-quality GNSS signal reception.

GAIN ROLL-OFF – Gain roll-off describes how the antenna's gain decreases as the elevation angle lowers from zenith to the horizon. It is determined by the antenna's beamwidth.

ANTENNA BEAMWIDTH – A wider beamwidth allows the antenna to receive more satellites across the sky, including those near the horizon, while a narrower beamwidth focuses energy more strongly toward the zenith.

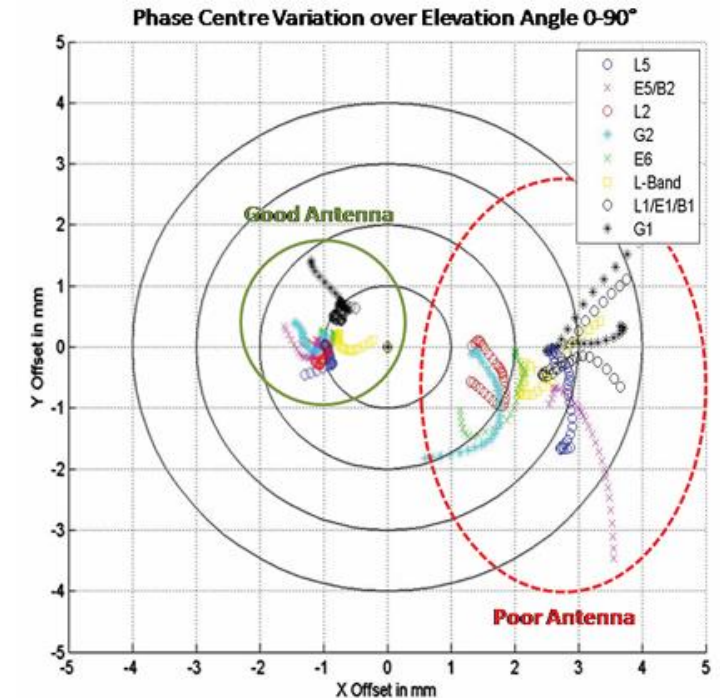
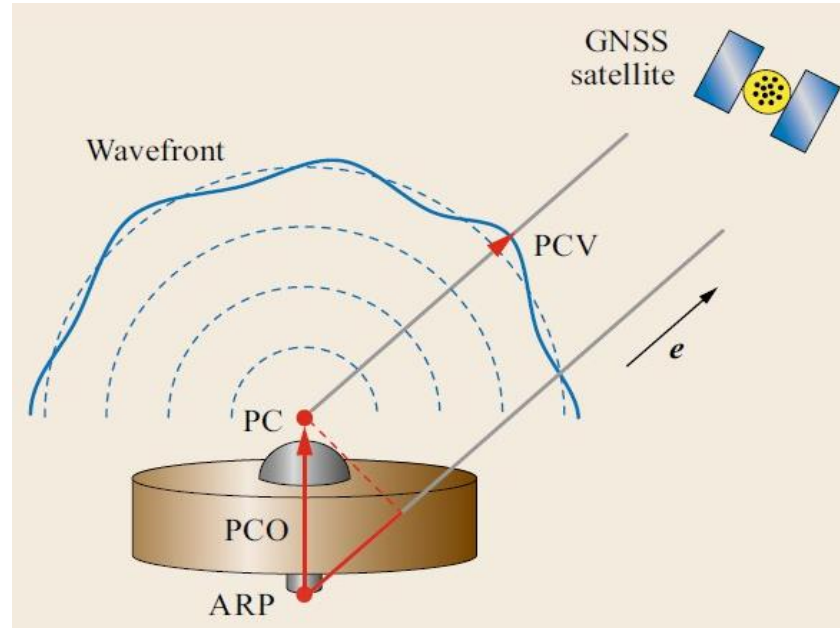
Bandwidth - It is the range of frequencies that a GNSS receiver's signal processing can accept and process. The receiver chooses a balance wide enough to get signal detail, narrow enough to limit noise.

Important Parameters for Choosing a GNSS Antenna

-Continued

- **PHASE CENTRE STABILITY-**

The phase centre of the antenna is the point where the signals transmitted from satellites are collected. Geodetic-grade antennas are used in survey applications because they offer superior control of PCO and PCV, ensuring stable and precise positioning results.



Plot of good and poor antenna phase centre variation over elevation angle

Source: Introduction to GNSS-NovAtel

Important Parameters for Choosing a GNSS Antenna

-Continued

- **APPLICATION -** A GNSS antenna must satisfy the performance, environmental, mechanical, and operational requirements of its intended application.

Antennas used in Survey

Antennas used in Aviation

Why GNSS Uses Circular Polarization?

- I. GNSS satellites transmit Circularly Polarized (CP) radio waves because CP significantly improves polarization efficiency at the receiving antenna.
- II. By using Right-Hand Circular Polarization (RHCP) at both the satellite and the receiver, polarization alignment is not required.
- III. CP waves mitigates the effects of Faraday rotation in the ionosphere ensuring more reliable reception.
- IV. CP antennas help reject multipath.

Types of GNSS Antenna

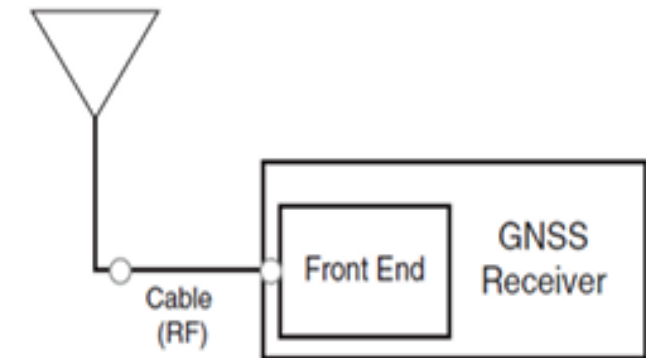
Passive Antenna

A passive GNSS antenna is an antenna that does not contain an internal amplifier. It receives GNSS signals directly from satellites and passes them to the receiver without any active electronic amplification.

Long cable not used (signal loss cannot be compensated)

Usage Handheld GNSS units
 Small consumer GPS devices (phones, trackers)
 UAV/Drone antennas
 Short cable installations

Passive Antenna



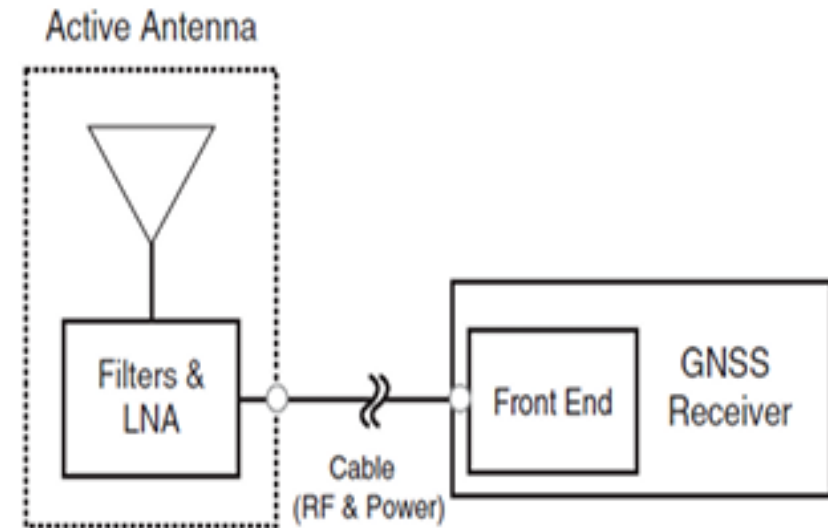
Types of GNSS Antenna

Active Antenna

An **active GNSS antenna** is an antenna that contains a **Band Pass Filter** and an internal **Low-Noise Amplifier (LNA)** to boost the very weak GNSS signals before they travel through the antenna cable to the receiver.

Supports long cable lengths running from the antenna to the receiver.

Usage Survey-grade GNSS systems
 CORS/reference stations
 Geodetic antennas (choke-ring, geodetic patch)
 External fixed installations with long cables



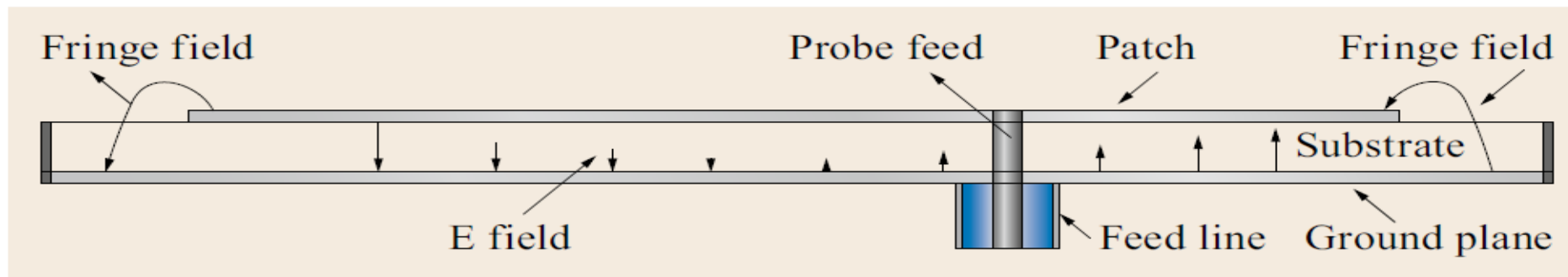
Types of GNSS Antenna

-Continued

Microstrip Patch Antenna (MPA):

- Compact, low-profile design widely used in mass-market and many professional receivers.
- Offers simple fabrication and can be stacked for multi-band operation, but has narrower bandwidth compared to some other designs.

Usage: Smartphones, UAVs/drones, Automotive GNSS, Handheld and portable GNSS receivers

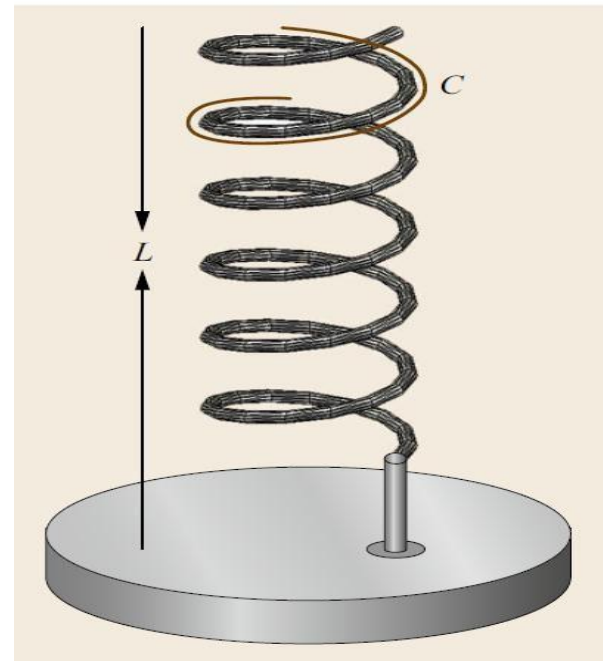


Types of GNSS Antenna

-Continued

Helix Antenna:

- Antenna consists of a wire shaped in the form of a helix with longitudinal axis perpendicular to ground.
- Popular for GNSS satellite antenna

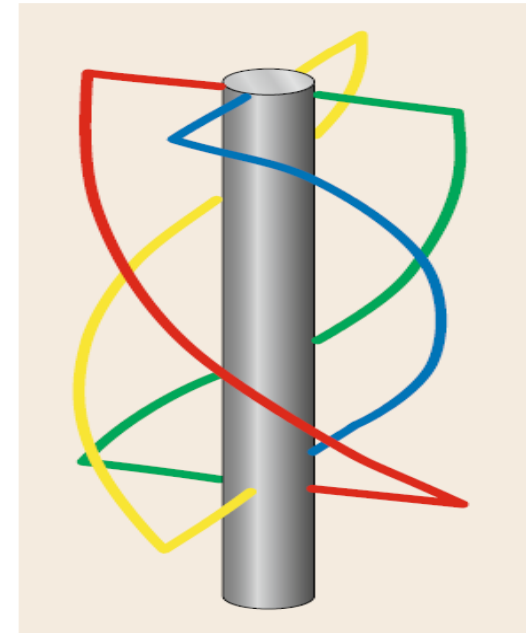


Types of GNSS Antenna

-Continued

Quadrifilar Helix Antenna (QHA):

- Compact CP antenna using four helices with progressive phasing (0, 90, 180, 270 degrees).
- QHA antennas are used in applications that require compact size, omnidirectional coverage, and good GNSS reception even when the device changes orientation.

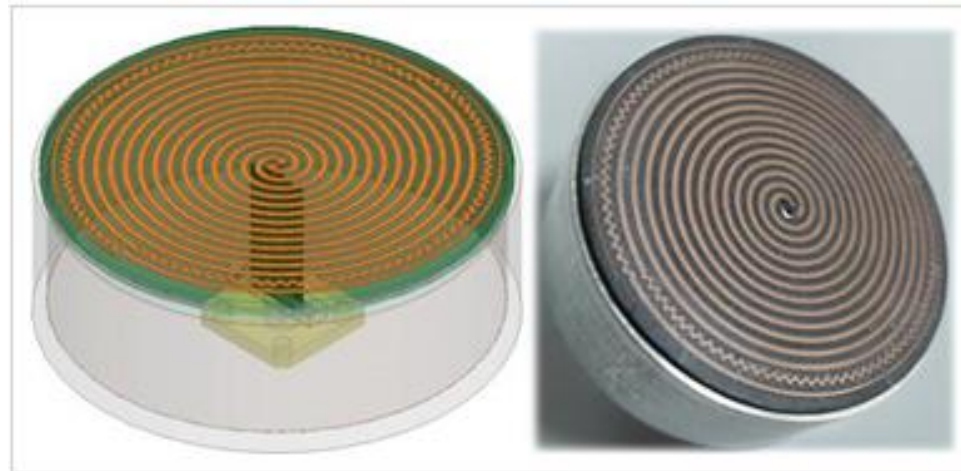


Types of GNSS Antenna

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Spiral antenna:

Spiral antennas are wideband, circularly-polarized antennas used in GNSS, satellite communication, direction-finding, radar, and aerospace applications due to their broad frequency range and stable performance.



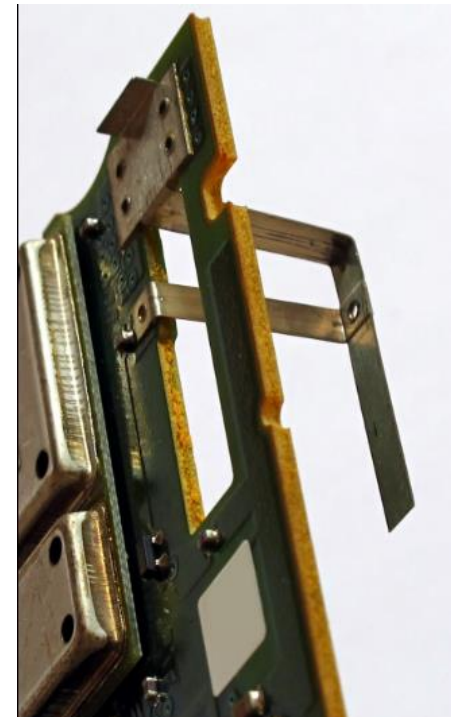
Application Specific GNSS Antenna

-Continued

Inverted F Antenna (IFA) :

Common in mobile phones; compact and typically linearly polarized. Suitable for consumer devices requiring moderate positioning performance.

An inverted-F antenna used for cordless phones and similar devices



Application Specific GNSS Antenna

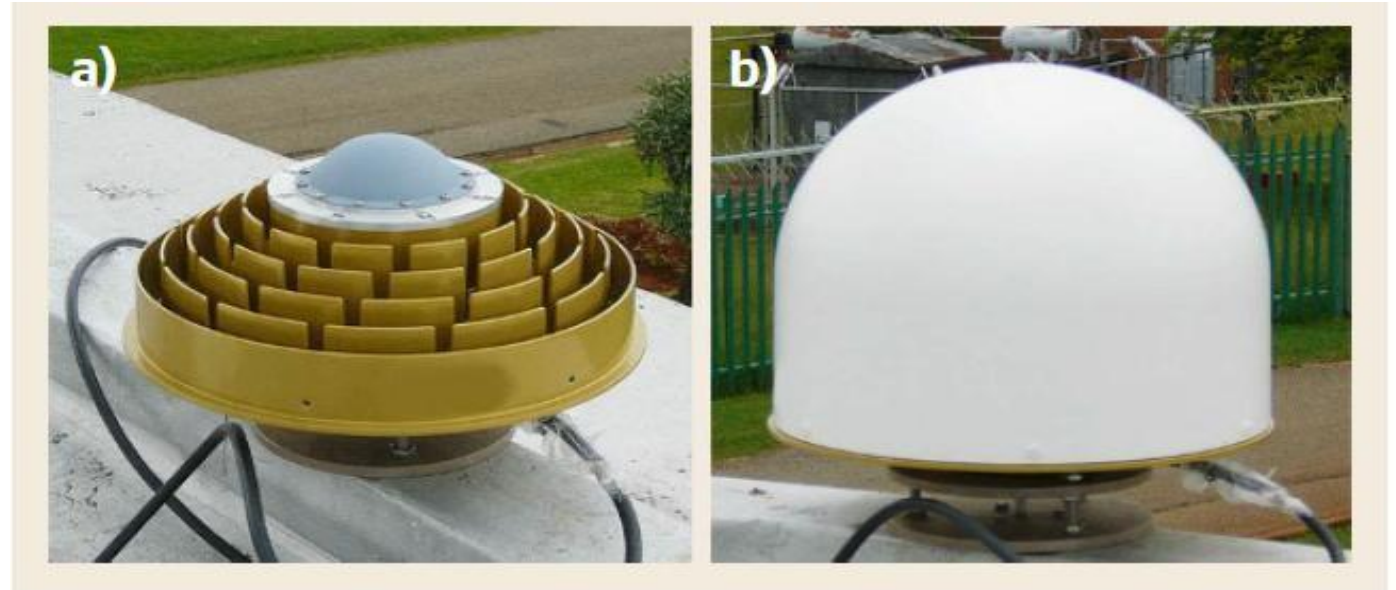
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Choke Ring Antenna:

Used for high-precision Survey, Geodesy and permanent CORS stations. Excellent at rejecting low elevation multipath and providing phase centre stability.

Dual or multi-frequency measurements.

Radome protects against rain and snowfall.



Application Specific GNSS Antenna

-Continued

Aviation:

Robust to sustain vibration and wide temperature range.

Low profile, conformal design for accommodation on aircraft to minimize drag and turbulence.

Antenna has Radome and tolerant to lightning.



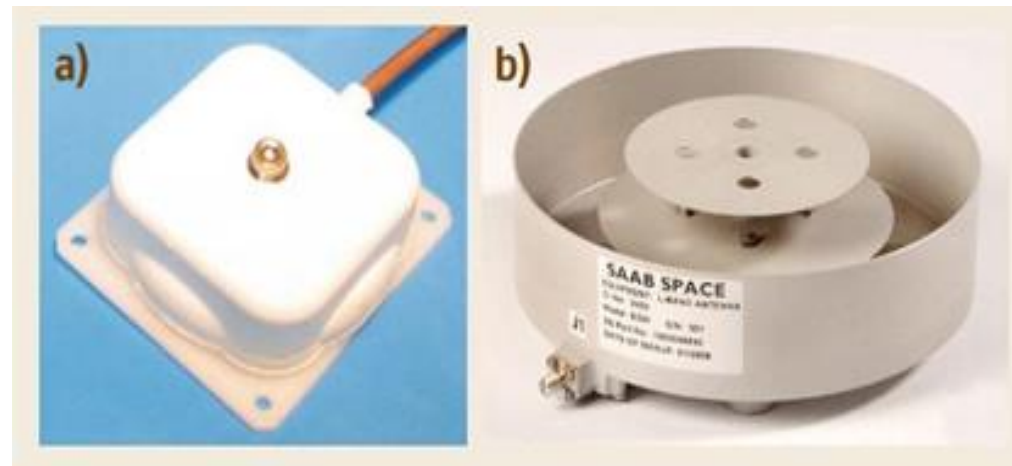
Application Specific GNSS Antenna

-Continued

Space Application:

GNSS antenna used on LEO satellite for PNT and precise orbit determination.

Such antennas meet qualification requirements of space (e.g. vacuum, compact, low weight, extreme temperature, shock-vibration, space radiation, low out-gassing material etc.)



Application Specific GNSS Antenna

-Continued

GNSS Remote Sensing & Reflectometry

Reflectometry uses the reflected GNSS signals (typically LHCP after reflection); antennas for this purpose may require larger off nadir gain and LHCP sensitivity.



Application Specific GNSS Antenna

-Continued

Anti-Jamming / Critical Infrastructure :

To protect GNSS signals from jamming, special antennas like CRPA (Controlled Radiation Pattern Antenna) or other mitigation techniques are used. These help ensure that important services depending on Position, Navigation, and Timing (PNT) continue to work even when interference is present.



GAJT-710MS anti-jam antenna | NovAtel

Thank You